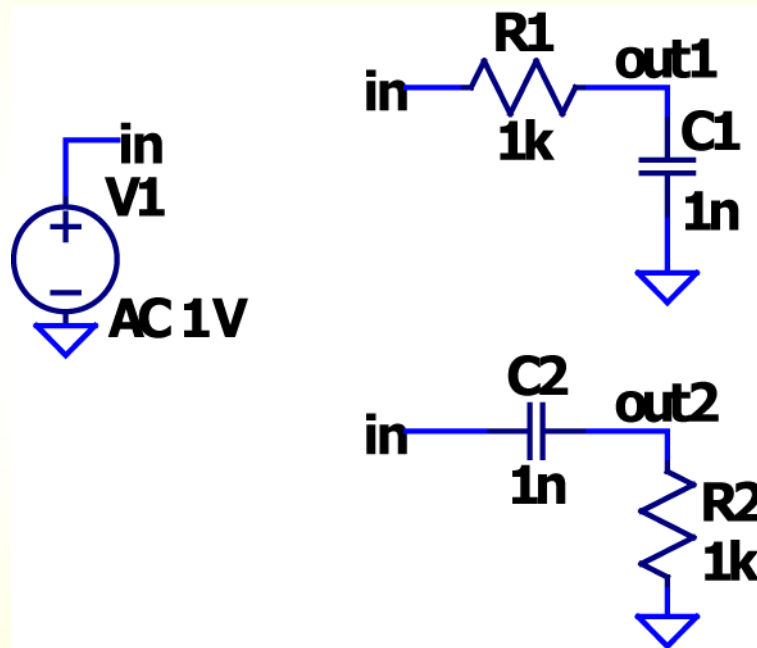




RC circuits

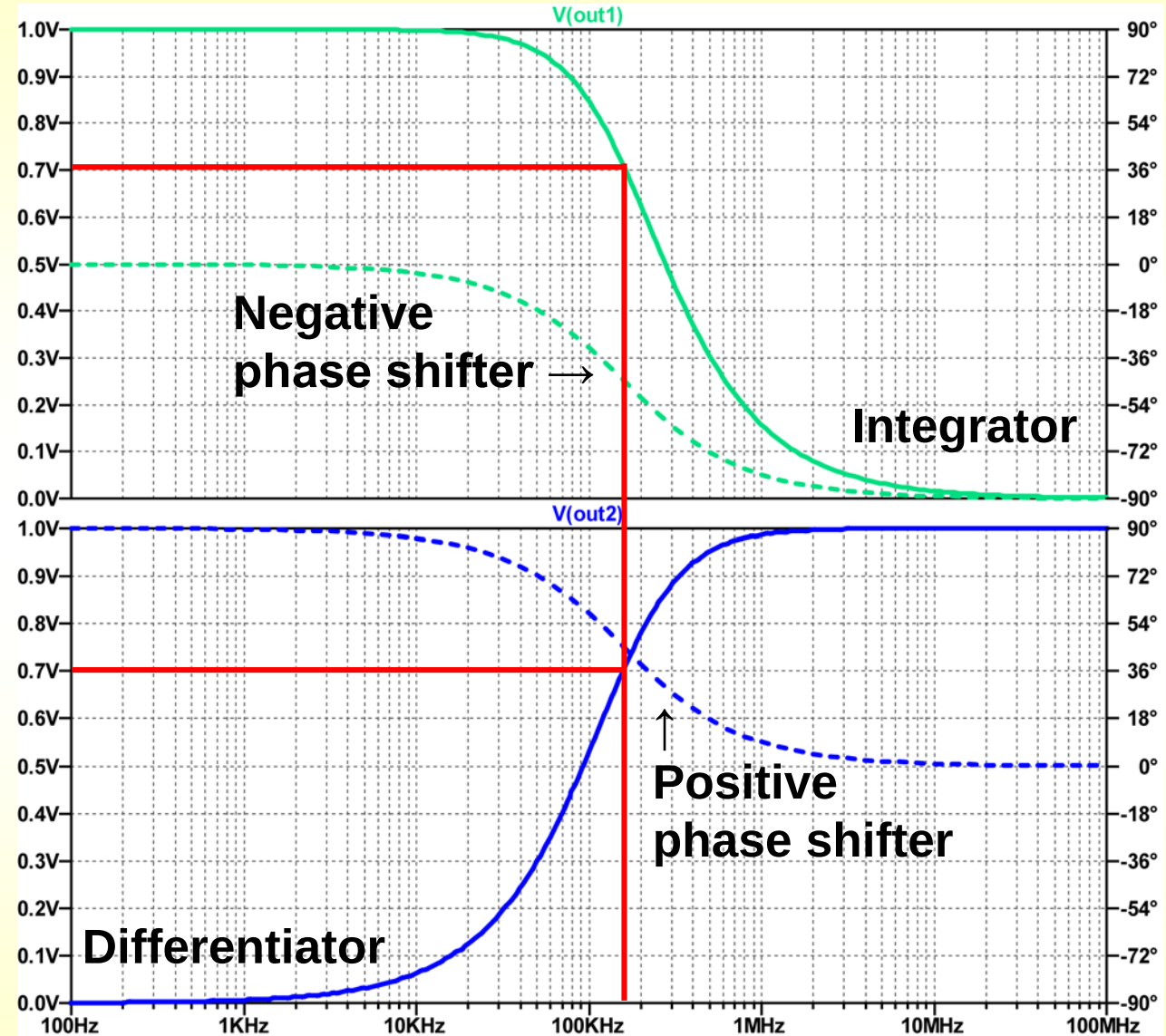
- Voltage output at C or R



$$f_B = \frac{1}{2\pi RC} = 159 \text{ kHz}$$

Low-pass
filter

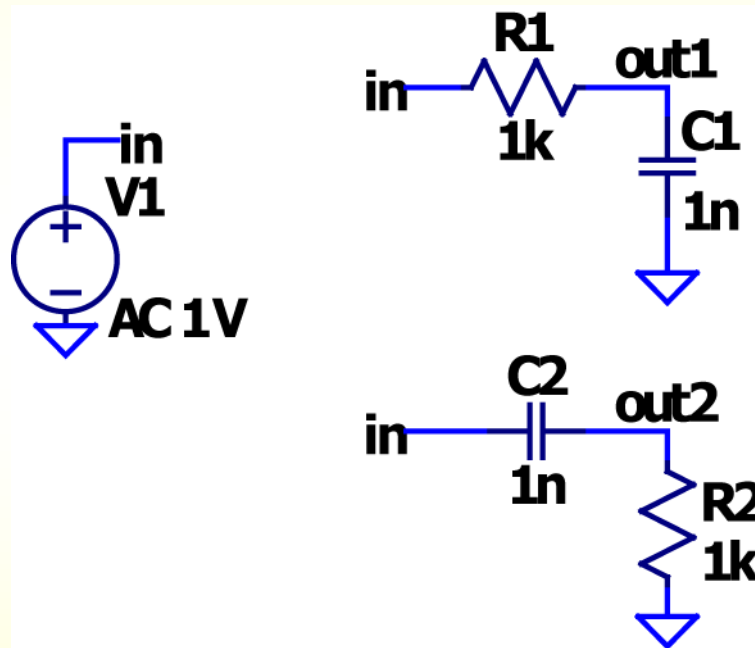
High-pass
filter





RC circuits

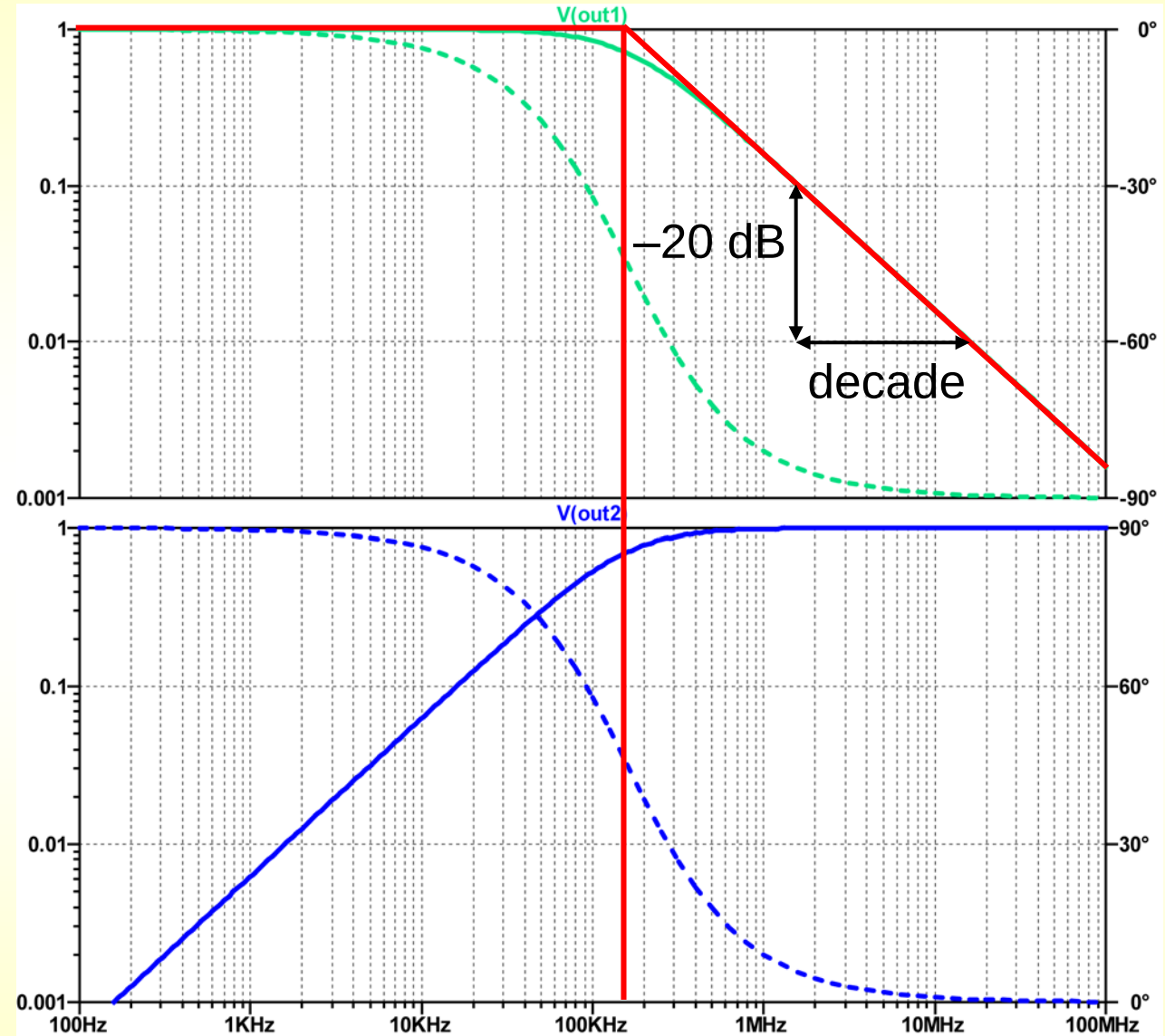
- Voltage output at C or R



Low-pass
filter

High-pass
filter

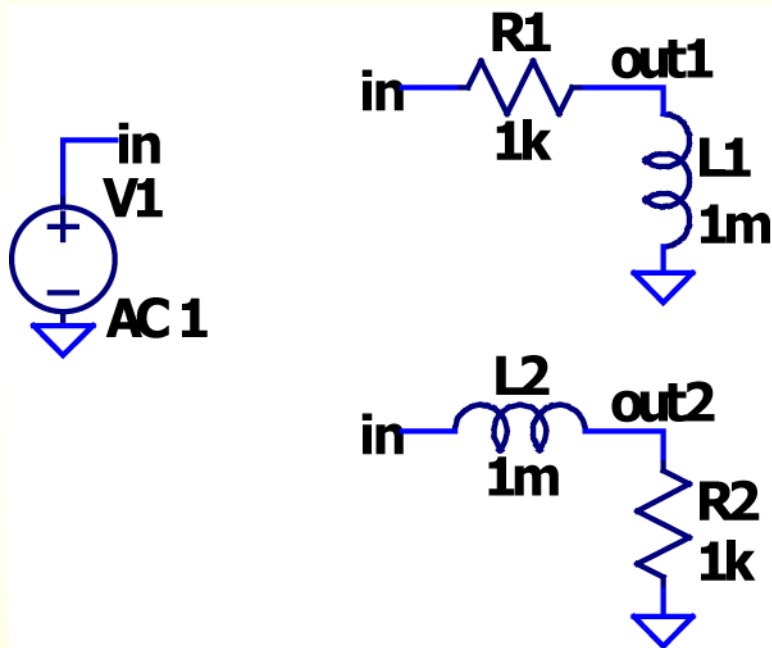
- Log-log plot
- Bode plot:** dB scale with straight lines





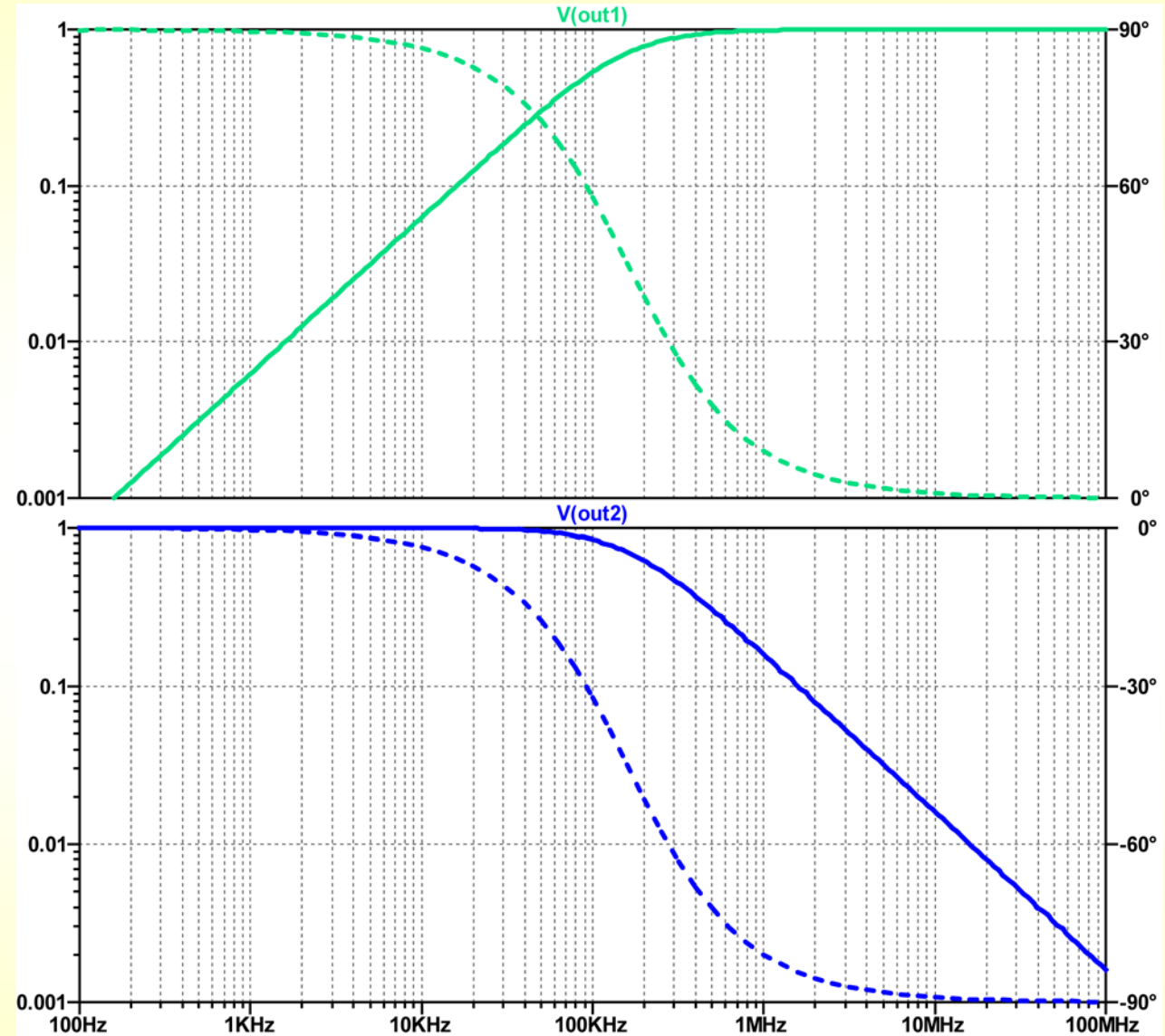
RL circuits

- Voltage output at L or R



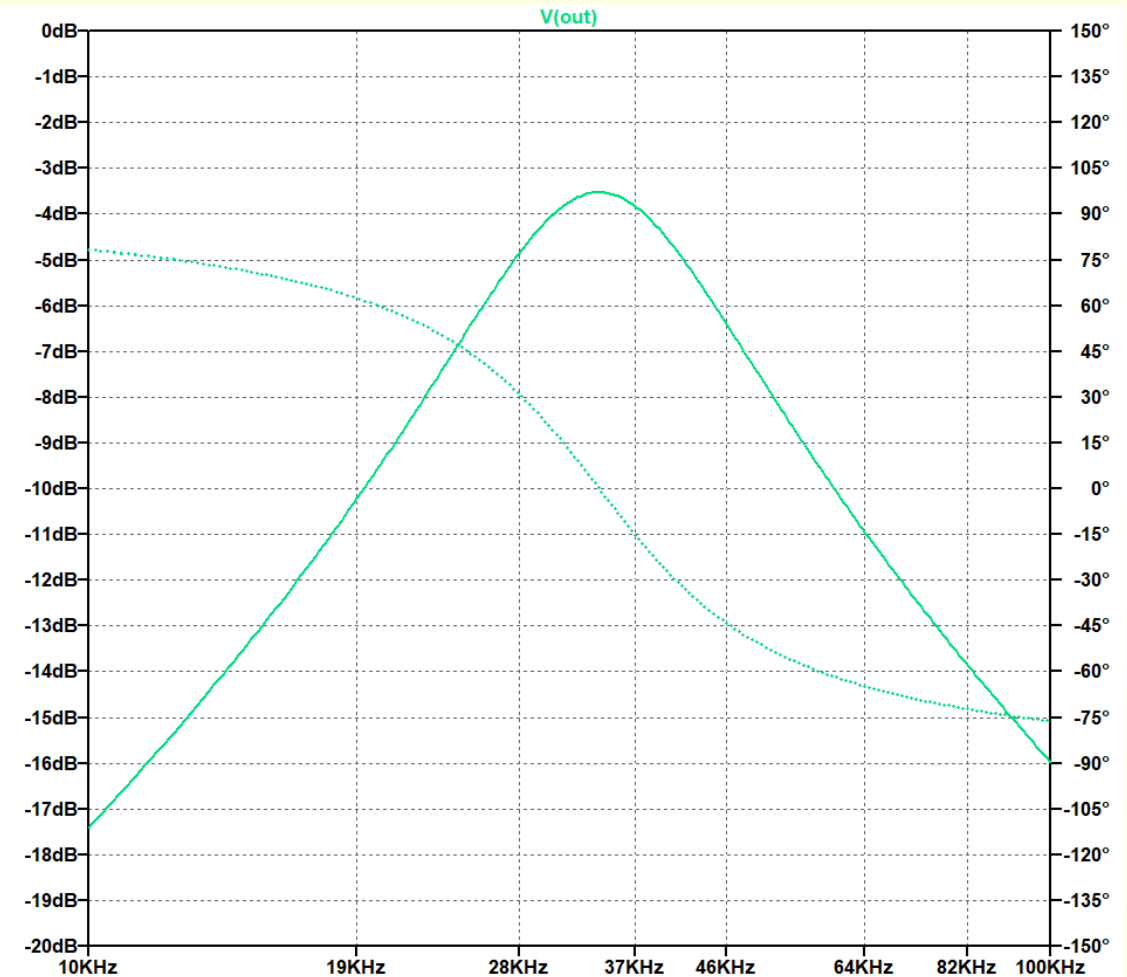
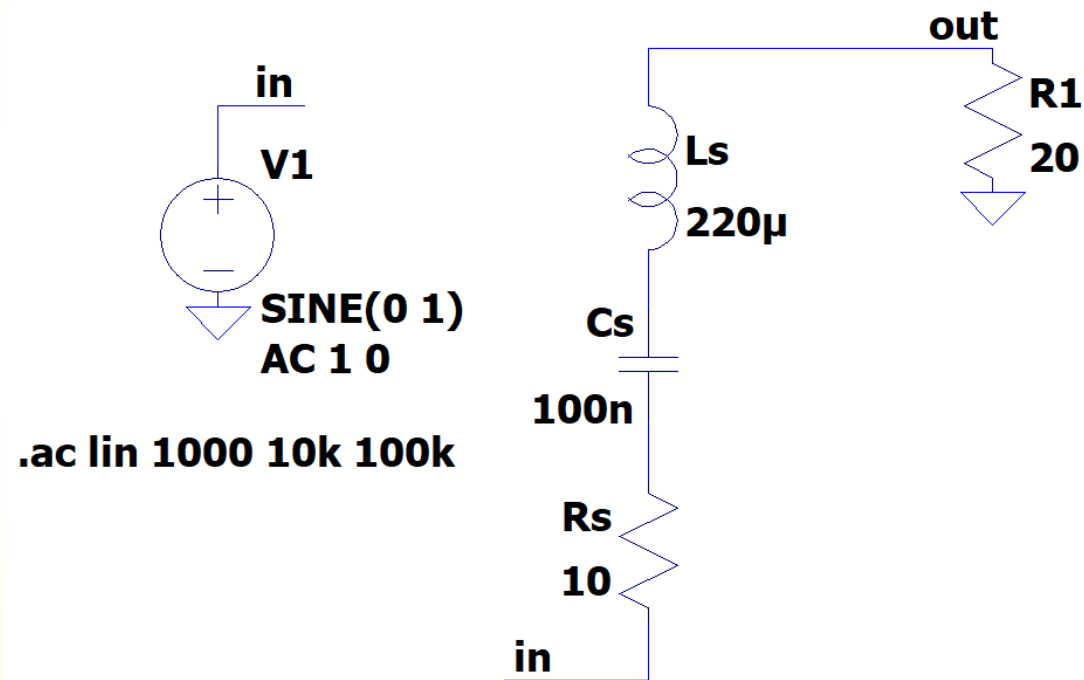
High-pass
filter

Low-pass
filter



Series RLC circuit

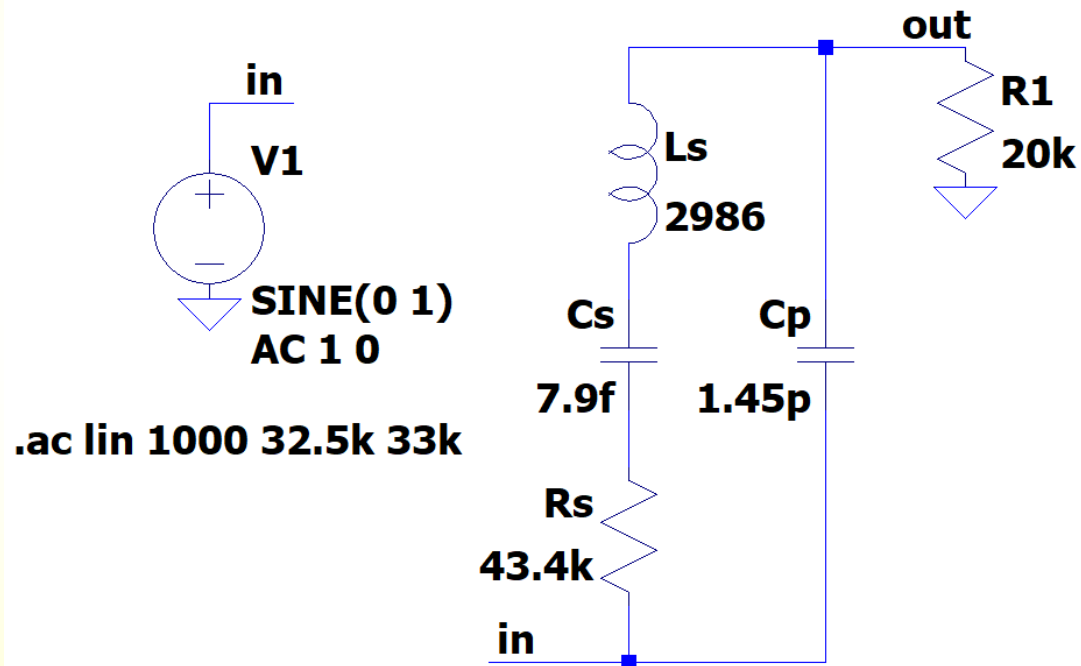
- Voltage output at R1



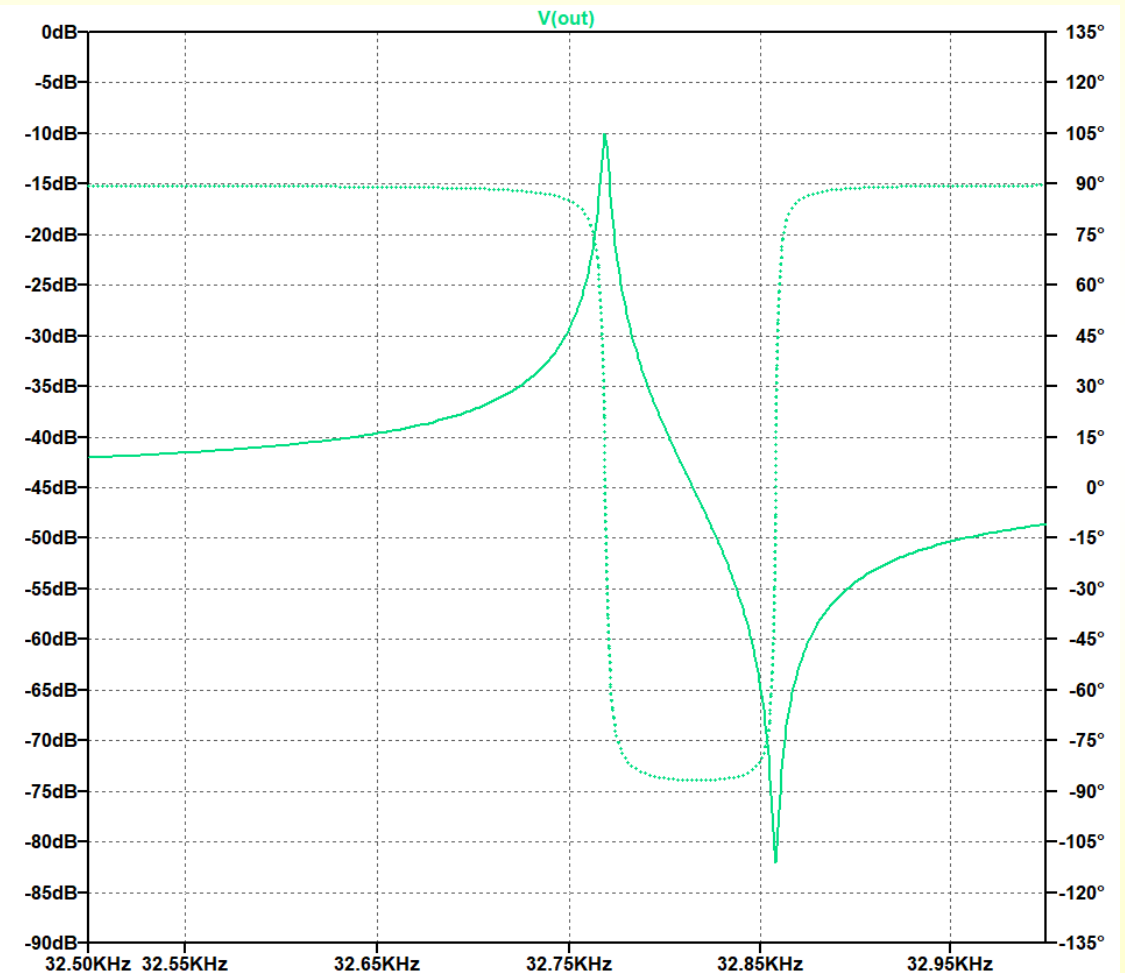


RLC model of a quartz oscillator

- Voltage output at R1 at series resonance



(32768.0 Hz nominal)





Harmonic analysis

- Square wave of fundamental + 3 harmonics

$$f_1 := 90 \text{ Hz} \quad \omega_1 := 2 \cdot \pi \cdot f_1 \quad V_0 := 1 \text{ V}$$

$$v_{sq}(t) := \frac{4}{\pi} \cdot V_0 \cdot \left(\sin(\omega_1 \cdot t) + \frac{1}{3} \cdot \sin(3 \cdot \omega_1 \cdot t) + \frac{1}{5} \cdot \sin(5 \cdot \omega_1 \cdot t) + \frac{1}{7} \cdot \sin(7 \cdot \omega_1 \cdot t) \right)$$

